

JavaScript / ECMAScript Exam Preparation

This document contains multiple choice questions, fill in the blanks, open questions, and answers for JavaScript, DOM, Events, and ECMAScript topics.

Part 1 — Multiple Choice Questions

1. What does `querySelector()` return?

- A) All matching elements
- B) First matching element
- C) Only div elements
- D) HTML collection

2. Which keyword should be used by default in modern JavaScript?

- A) var
- B) static
- C) const
- D) public

3. Which method converts an object into JSON string?

- A) `JSON.parse()`
- B) `JSON.stringify()`
- C) `toJSON()`
- D) `parseJSON()`

4. What does `classList.toggle()` do?

- A) Deletes all classes
- B) Changes CSS file
- C) Adds/removes class
- D) Hides element

5. Which event is commonly used for keyboard input?

- A) `mousemove`
- B) `click`
- C) `keydown`
- D) `submit`

6. What does `preventDefault()` do?

- A) Deletes event
- B) Prevents default browser behavior
- C) Stops JS execution

D) Removes HTML element

7. Which method is used to attach events?

- A) querySelector()
- B) createElement()
- C) addEventListener()
- D) appendChild()

8. Which method filters array elements?

- A) map()
- B) reduce()
- C) filter()
- D) push()

9. Which ES version introduced async/await?

- A) ES5
- B) ES6
- C) ES2017
- D) ES2025

10. Which ES version introduced classes?

- A) ES5
- B) ES6
- C) ES2020
- D) ES2025

Part 2 — Fill in the Blanks

1. `const el = document._____("#title");`
2. `el.classList._____("active");`
3. `const div = document._____("div");`
4. `parent._____(child);`
5. `btn._____("click", handleClick);`
6. `const json = JSON._____(user);`
7. `const obj = JSON._____(json);`
8. `const text = input._____;`
9. `e._____();`
10. `const doubled = nums._____(n => n * 2);`

11. `const evens = nums._____ (n => n % 2 === 0);`
12. `const sum = nums._____ ((acc, n) => acc + n, 0);`
13. `window.location._____ = "profile.html";`
14. ES6 was released in the year _____.
15. Optional chaining operator is written as _____.

Part 3 — Open Questions

1. Explain the difference between `textContent` and `innerHTML`.
2. Explain `appendChild()` and `createElement()`.
3. Why is `const` preferred over `var`?
4. Explain how `classList.toggle()` works.
5. Describe the flow of adding a new task in a `ToDo` app.
6. Explain `querySelector` and `querySelectorAll`.
7. What is JSON and why is it important in web development?
8. Explain `map`, `filter`, and `reduce`.
9. What is DOM?
10. What happens when `window.location.href` changes?

Answers

Multiple Choice Answers

1. B
2. C
3. B
4. C
5. C
6. B
7. C
8. C
9. C
10. B

Fill in the Blanks Answers

1. `querySelector`
2. `add`
3. `createElement`
4. `appendChild`
5. `addEventListener`
6. `stringify`
7. `parse`
8. `value`
9. `preventDefault`
10. `map`
11. `filter`
12. `reduce`
13. `href`
14. 2015
15. `?`.

Open Question Key Points

- `textContent` inserts plain text safely, while `innerHTML` parses HTML.
- `createElement` creates a new HTML element; `appendChild` adds it into the DOM.
- `const` is block-scoped and safer than `var` because `var` leaks scope and hoists.
- `toggle()` adds a class if absent and removes it if present.

- Read `input.value`, `createElement('li')`, `appendChild()`, `addEventListener()`.
- `querySelector` returns first match; `querySelectorAll` returns all matches.
- JSON is the standard data exchange format between client and server.
- `map` transforms elements, `filter` keeps matching elements, `reduce` combines values.
- DOM is the Document Object Model — tree representation of HTML.
- `window.location.href` changes the current page URL and redirects the user.